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Tripartite Microbial Community Dynamics:

A Theoretical Framework for Simulating Competitive and Cooperative Strategies

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This study presents a mathematical framework for investigating the temporal dynamics of co-existence and competition among heterotrophic microbes within complex communities on short and longer time scales. Focusing on metabolic interactions, it explores how distinct heterotrophic strategies: public metabolizing, private metabolizing, and cheating, shape population behaviour. The framework integrates generalized Lotka-Volterra dynamics with an evolutionary game theory (EGT) to capture the effects of resource exchange, particularly the release of glucose by public metabolizers and the availability of sucrose as a shared substrate that drives population growth. In our model, EGT allows us to explore the frequency-dependent dynamics among three microbial strategies by assigning payoffs that reflect their ecological costs and benefits. A robust numerical analysis reveals the conditions under which cooperation is stabilized and identifies ecological thresholds for the survival of a synthetic, cross-kingdom system linked by nutrient exchange. These insights offer general principles for microbial co-existence and may inform future efforts in ecosystem engineering, biocontrol, and synthetic community engineering.

1. Introduction

Microorganisms are omnipresent in nature, where they form diverse communities composed of genetically different strains or even a variety of species that interact and co-exist. For example, the human colon microbiota includes 100-300 different bacterial taxa, along with dozens of fungi and viruses [1]. Cloud water samples

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collected at high altitudes revealed over 28,000 unique types of bacteria and approximately 2,600 eukaryotic species [2], while soil microbial consortia, composed of bacteria, fungi, and archaea, represent a substantial part of Earth's biodiversity [3]. These natural communities play a vital role in biogeochemical processes that directly influence soil quality and enhance primary productivity [4]. Alongside efforts to understand natural microbiomes, there is increasing interest in the rational design of synthetic microbial communities, with defined composition and controlled interactions, for targeted applications [5]. Engineered consortia have been developed for a variety of biotechnological and medical purposes, including, among many, gut microbiome therapies [6], bioremediation, industrial bioproduction of biofuels, pharmaceuticals, and enzymes [7,8].

A key feature of microbial communities, both natural and synthetic, is their capacity to release a wide range of compounds into their surroundings. The constituent microbes secrete metabolites, enzymes, and signalling molecules into their environment, shaping the ecological landscape [9,10]. These secretions often serve as the basis for cooperative, competitive, or neutral interactions [10]. For example, in mutualistic cross-feeding, the fungus *Laccaria bicolor* secretes trehalose to attract the bacterium *Pseudomonas aeruginosa*, which in turn produces thiamine to support fungal growth [11]. Some strains of *Saccharomyces cerevisiae* (baker's yeast) secrete invertase into the periplasmic space or environment, allowing sucrose to be hydrolyzed outside the cell [12,13]. This sugar then becomes accessible to both the producing and neighbouring non-producing cells.

To better understand how such interactions govern community dynamics and long-term stability, one may adopt an economic perspective, drawing analogies between microbial exchange systems and resource trade in human markets. Much like market agents, microbes exchange resources to survive and grow. The trade of nutrients and metabolites is similar to transactions in human economies, where the exchange of goods such as food, energy, or raw materials affects relationships, community behaviour, and overall outcomes [14]. Cooperation within these systems emerges when certain microbes produce and share public goods, such as extracellular enzymes [15,16]. This behaviour is comparable to businesses investing in shared infrastructure. Yet, as in human economies, competition also arises: Some microbes act as free riders, exploiting communal resources without contributing, thereby destabilizing cooperative systems. To reflect this perspective, a nomenclature inspired by economics is adopted throughout the paper.

Focusing on the metabolic strategies reveals the diverse roles that microbes can play in shared environments. Some microbes engage in mutualistic interactions by releasing extracellular enzymes or metabolites that benefit others, fostering cooperative relationships that can enhance the entire community [17]. In the literature, these organisms are often referred to as cooperators [18] or public good producers [19]. In this paper, we refer to them as public metabolizers (P). However, public metabolizer populations are often vulnerable to exploitation by cheaters (C), individuals who avoid the energetic cost of enzyme production, but still reap the benefits of the communal pool of resources [20,21]. Cheating, also referred to as free riding [22] or defecting [12,13], has been widely studied as a form of negative interaction, introducing competition that can disrupt community balance [23]. By taking advantage of shared resources without contributing to their maintenance, cheaters add complexity to microbial interactions, straining cooperative systems and intensifying competition. The interplay between public metabolizers and cheaters influences not only the fitness of individual organisms but also the overall balance and ecological stability of microbial ecosystems. For instance, cheating can destabilize cooperative systems by reducing the net benefit of public goods production, leading to population crashes or oscillations [20,21,23,24]. Somewhere in between are the properties of private metabolizers (M): neither cooperative nor openly parasitic. They import shared resources from the environment and digest them internally, keeping enzymes such as invertase within the cell [15,20]. While they do not contribute to the communal pool of public goods and thus do not support cheaters, they also bear the metabolic cost of enzyme production, distinguishing them from pure exploiters. Ecological studies combined with mathematical modelling reveal that private metabolizers can invade and replace an initially stable group composed of public metabolizers and cheaters [15]. Although neutral toward cheaters due to their lack of public resource sharing, private

metabolizers directly compete with public metabolizers for the same environmental substrates, thereby influencing community structure and stability [20].

Although extensive research has been devoted to this topic, the mechanisms shaping community composition and species interactions remain only partially understood. Mathematical modelling offers a powerful framework to explore these dynamics systematically. Building on foundational approaches such as the Lotka–Volterra predator–prey models [25,26], ordinary differential equations (ODEs) have become central tools in studying population dynamics across ecological and microbial systems [27,28]. The application of mathematical and computational models in biology and ecology has a rich history, tracing back to studies in the early 20th century [29]. In the context of this work, such models help elucidate the consequences of metabolic strategies in shared-resource environments. By capturing how population sizes change over time, ODE-based models allow us to explore the conditions under which cooperation, cheating, and privatization can persist or collapse, providing insights that are often inaccessible through empirical work alone.

While ODE-based ecological models capture short- to medium-term population dynamics, understanding the long-term outcomes of these interactions requires an evolutionary perspective. Laboratory evolution experiments, in which microbial populations are allowed to evolve over many generations, provide useful insights. Coupling such empirical approaches with evolutionary game theory (EGT) [30,31] allows for dynamic modelling of strategy changes across generations, where offspring may adopt different behaviours based on the relative fitness of competing types [32]. In EGT, biological agents take the place of rational players in classical game theory, and natural selection, rather than explicit decision-making, governs reproductive success [33,34]. A major conceptual advance came in 1973 with John Maynard Smith's introduction of the evolutionarily stable strategy (ESS), which extended the Nash equilibrium to biological systems and provided a formalism for studying frequency-dependent selection [31,35]. With the arrival of the new millennium, researchers began adapting game theory frameworks to microbial systems, particularly to understand cooperation and the evolution of virulence. The field gained momentum in the 2000s with a surge of studies applying EGT to microbial cooperation and cheating [13,18,36]. A key example is the work of Rainey and colleagues, who demonstrated how social cheaters could invade and destabilize cooperative bacterial populations [37], illustrating how game-theoretic models can explain the evolutionary dynamics underlying microbial social behaviour.

In this study, we develop a tripartite computational model to explore the ecological and evolutionary dynamics among three microbial strategies: public metabolizers, private metabolizers, and cheaters. The model builds on a generalized Lotka–Volterra framework [38], where population growth is influenced by access to resources, specifically glucose and sucrose (captured in Figure 1). Our main objective is to determine the conditions under which all three strategies can stably coexist.

Prior studies have shown that in systems composed solely of public and cheating strains, population levels often undergo continuous oscillations [39]. However, the introduction of a private metabolizer fundamentally alters these dynamics: private strains can dominate the system, driving both public metabolizers and cheaters to extinction [15]. This outcome underscores the competitive edge of metabolic privatization and the fragility of cooperative networks facing exploitation.

To incorporate the evolutionary dimension, we extended our analysis using EGT, enabling strategy shifts across generations based on relative fitness. This approach extends earlier game-theoretical analyses [12,13] of a related system by including the type of private metabolizer. It reveals not only which populations persist, but also which strategies are evolutionarily stable in the long term. By integrating ecological dynamics with evolutionary feedbacks, we capture a fuller picture of how microbial communities adapt and reorganize in response to internal and external pressures. Beyond theoretical insights, our findings have practical implications for the design and management of synthetic microbial consortia. Understanding how cooperation

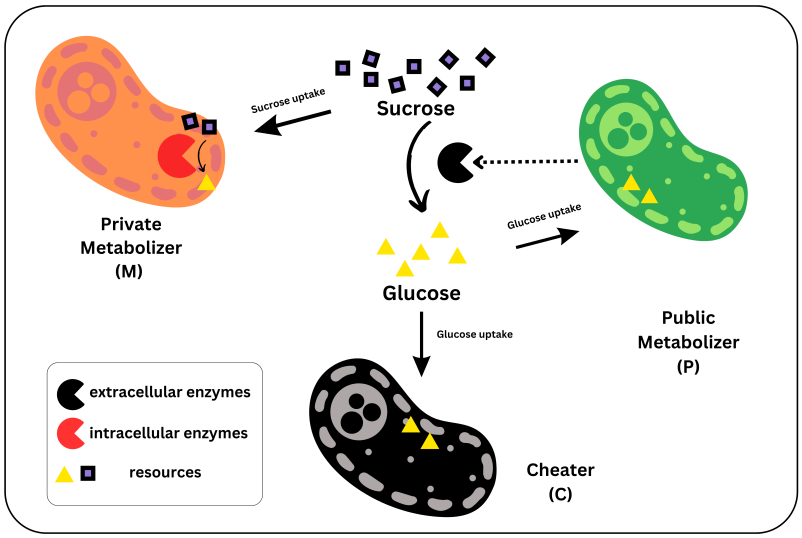

Figure 1. Schematic representation of a tripartite microbial community, captured based on their sugar metabolism. All organisms are heterotrophic and utilize glucose (yellow triangle) derived from the breakdown of sucrose (black square). Public metabolizers (P) secrete extracellular enzymes (black Pac-Man symbol) to hydrolyse sucrose into glucose, contributing to a public pool of accessible resources. Cheaters (C) do not produce enzymes but exploit the publicly available glucose without incurring production costs. Private metabolizers (M) import sucrose for intracellular digestion using internal enzymes (red Pac-Man symbol), incurring metabolic costs while avoiding resource sharing. This scheme forms the conceptual basis for a mathematical model exploring the ecological dynamics of cooperation, exploitation, and metabolic privatization in microbial communities.

can be maintained in the presence of cheaters supports efforts in biotechnology, environmental engineering, and medicine [13,19,40].

2. Modelling framework

Through this work, we model a closed, well-mixed batch culture, where three microbial strategies can be adopted: a public metabolizer (P), a cheater (C), and a private metabolizer (M) (Figure 1). Sucrose is continuously supplied as the sole carbon source. The public metabolizer secretes extracellular invertase, which hydrolyzes sucrose into glucose in the surrounding medium. This glucose is partially assimilated for its growth, while the rest becomes part of a common resource pool accessible to other community members.

(a) Dynamic model of the tripartite population

To describe the ecological dynamics of these three strategies, we employ a generalized Lotka–Volterra framework that captures intrinsic growth, pairwise interactions, and density-dependent effects. Importantly, we assume that public metabolizers constitutively express invertase at a constant rate throughout their lifetime. Consequently, we do not explicitly model the dynamics of enzyme regulation, as done in other approaches [20,23,41], reducing the number of free parameters. At the same time, it preserves ecological realism, since cheater growth remains directly dependent on the activity and density of public metabolizers.

The resulting system is governed by a set of coupled ODEs:

$$\begin{cases} \frac{dP}{dt} & = r_p \cdot P - \alpha \cdot P \cdot C - \beta \cdot P \cdot M - \eta \cdot P^2 \\ \frac{dC}{dt} & = \alpha \cdot P \cdot C - \nu \cdot C^2 \\ \frac{dM}{dt} & = r_m \cdot M - \beta \cdot M \cdot P - \gamma \cdot M^2 \end{cases} \tag{2.1}$$

with seven parameters: intrinsic growth rates (r_p, r_m), cooperation (α), competition (β), and density-dependent constraints (η,γ, ν). The cheater population does not contribute to enzyme production and hence relies entirely on the glucose released by the public metabolizers. Its growth rate is therefore jointly determined by the cooperation coefficient α and the density of public metabolizers. However, to prevent uncontrolled proliferation, cheater growth is also regulated by density-dependent constraints ν. This model is designed to be readily parameterizable from empirical data. To ensure broad applicability and experimental integration, we present a stepwise protocol for parameter estimation based on monoculture and pairwise co-culture experiments. This framework enables researchers to directly map ecological interactions onto measurable population dynamics.

(i) Model parameterization

Intrinsic growth rates of P (r_p) and M (r_m) can be estimated from exponential growth phase data of monoculture experiments. As population densities increase, density-dependent constraints (η and γ) capture the effect of crowding and resource limitation near the carrying capacity (Figure S2). These parameters are obtained by fitting the following simplified equations:

$$\begin{cases} \frac{dP}{dt} & = r_p \cdot P - \eta \cdot P^2 \\ \frac{dM}{dt} & = r_m \cdot M - \gamma \cdot M^2 \end{cases} \tag{2.2}$$

To further parameterize the interaction terms, we can analyze data from a pairwise co-culturing experiment, allowing us to isolate and quantify key interaction coefficients that define inter-strategy dynamics. The exploitation coefficient α is estimated by comparing the growth of the P in monoculture and in co-culture with C (Eq.2.3):

$$\begin{cases} \frac{dP}{dt} = r_p \cdot P - \alpha \cdot P \cdot C - \eta \cdot P^2 \\ \frac{dC}{dt} = \alpha \cdot P \cdot C - \nu \cdot C^2. \end{cases} \tag{2.3}$$

Similarly, the competition coefficient β quantifies interference between public and private metabolizers, both of which utilize the same external sucrose supply (Figure S2). Although they employ different processing strategies, this shared resource creates indirect competition. To estimate β, we compare the growth of M in monoculture to its dynamics in co-culture with P (Eq.2.4):

$$\begin{cases} \frac{dP}{dt} = r_p \cdot P - \beta \cdot P \cdot M - \eta \cdot P^2 \\ \frac{dM}{dt} = r_m \cdot M - \beta \cdot M \cdot P - \gamma \cdot M^2 \end{cases} \tag{2.4}$$

The three systems (2-1) - (2.3) provide minimalistic, yet interpretable, representations of microbial ecological dynamics, with clearly defined interaction terms and parameters grounded in empirical observables. It also lays the foundation for subsequent stability analyses, allowing us to explore co-existence regimes, dominance shifts, and resilience under varying environmental conditions.

(b) Evolutionary Game Theory and Microbial Strategies

The outcome of interactions of subpopulations exhibiting different metabolic strategies is shaped by natural selection and often depends on the frequency of competing strategies, which EGT captures effectively. Our study considers a three-strategy game, where each of the three mechanisms represents a distinct subpopulation. To assign payoffs appropriately, our model accounts for both the benefits and drawbacks experienced by each subgroup.

(i) Three-strategy evolutionary game

The payoff illustrates the complex relationships and fitness outcomes among these strategies within a microbiological community. From an evolutionary perspective, natural selection often favours cheaters, as they benefit from shared resources without contributing to their production. This typically leads to a decline in the P population. The success of each strategy is thus dependent on population densities and the payoff of that strategy [42]. For different strategies, player payoffs can be interpreted as follows:

i) Public Metabolizers:

$$F_p = r_p - \eta P - \alpha C - \beta M,$$

ii) Cheaters :

$$F_c = \alpha P - \nu C,$$

iii) Private Metabolizers :

$$F_m = r_m - \gamma M - \beta P,$$

In this model, payoff functions are directly mapped to the intrinsic growth rates of individual strategies, serving as quantitative measures of fitness that govern population dynamics. A strategy with a higher payoff exhibits a growth advantage, increasing in relative abundance over time. For instance, when public metabolizers receive greater payoffs, the system tends to favour their expansion, typically resulting in a community enriched in public producers and the cheaters that exploit them. Conversely, when private metabolizers outperform their counterparts, population dynamics shift toward private dominance, favouring self-sufficiency over cooperative contribution.

3. Results

To investigate the ecological and evolutionary dynamics of our tripartite microbial system, we first performed a series of numerical simulations based on a generalized Lotka–Volterra framework. This approach enabled us to systematically vary key parameters, such as intrinsic growth rates and interaction strengths, and examine how these shape population trajectories over time. We also performed a sensitivity analysis that quantifies the impact of perturbing key model parameters on steady-state outcomes. This approach enables us to identify which interactions have the strongest influence on system behaviour and long-term community structure.

As shown in Figure S4, an increase in α, which represents the cooperative benefit allocated from P to C, leads to a notable positive impact on the M population. This outcome suggests that when C extracts excessive benefits from the P, the cooperative mechanism becomes unstable, indirectly giving private metabolizers a competitive edge. In contrast, the intrinsic growth rate of the private metabolizer has a strong positive effect on its steady-state abundance, indicating that its population dynamics are primarily governed by its growth potential rather than by interactions with the public or cheater populations. Similarly, an increase in the public metabolizer's growth rate benefits both the P and C populations due to their mutual dependency while having minimal influence on the M population.

(a) Impact of the growth rate

We started the analysis of our model by testing various growth rates while keeping other parameters constant, thereby allowing us to test the dynamics of species interactions within a shared environment. In scenarios involving public and private metabolizers, stable coexistence is critically dependent on the balance between their respective growth rates (r_p and r_m). The public metabolizer plays a vital role in maintaining community function, potentially by producing shared resources or maintaining favourable environmental conditions that benefit the total community. However, when the growth rate of the private metabolizer becomes excessively high, it can dominate the system by consuming a disproportionate share of resources or suppressing

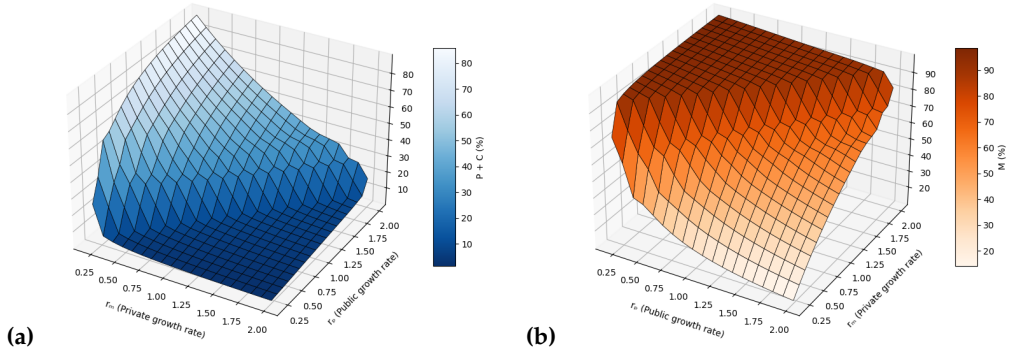

Figure 2. Impact of different growth conditions of P and M on population abundance. In this figure, we systematically investigate how changes in the intrinsic growth rate parameters r_p and r_m influence the ratio of community distribution. (a) P and C ratio in total population (b) M ratio in total population

the growth of the public metabolizer. This imbalance may result in system instability, and thereby leading to the extinction of both the public metabolizer and the cheater population (Figure 2a).

Numerical simulations demonstrate that stable coexistence among public, cheater, and private metabolizers is highly sensitive to the balance of their growth rates. In particular, the public metabolizer must sustain a growth rate sufficiently high to maintain its own population while also supporting the cheater population that relies on shared resources. However, this growth advantage must be carefully constrained: when the public metabolizer's growth rate exceeds approximately 2 times that of the private metabolizer, competitive pressures lead to the decline or extinction of the private metabolizer (Figure 2b). Conversely, if the public metabolizer's growth rate falls below roughly 1.2 times that of the private metabolizer, neither the public nor the cheater populations can persist, resulting in the dominance of the private metabolizer. Therefore, a narrowly defined growth rate ratio, between 1.2 and 2 times higher for public metabolizers, is essential to ensure balanced and stable coexistence among all three metabolic strategies.

Further analysis of population dynamics reveals that excessive growth by any single population eventually leads to overcrowding, resource depletion, and reduced viability across the entire community. Similarly, simulations show that in the absence of one population, the remaining species initially may experience optimal growth conditions. However, they soon encounter environmental constraints imposed by limited resources. These findings emphasize the fundamental roles of interspecies competition, resource availability, and ecological balance in shaping community structure and stability. Collectively, these results highlight the interdependence of ecological interactions and demonstrate how even minor variations in growth dynamics can substantially influence the persistence and composition of microbial communities.

(b) Impact of initial population size

The dynamic interplay among public metabolizers, cheaters, and private metabolizers in microbial communities reflects a finely balanced ecological system, significantly influenced by initial population densities in the short term, yet exhibiting long-term robustness under balanced growth conditions. During the early phases of community development, differences in initial population composition influence short-term dynamics. When cheaters are introduced at low initial densities, public metabolizers can sustain their growth while simultaneously supporting the cheater population through the provision of shared resources (Figure S1a). Conversely, elevated initial densities of cheaters accelerate resource depletion, prompting a temporary decline in public metabolizers and destabilizing the early stages of community structure (Figure S1b). Despite early fluctuations, our long-term simulations consistently demonstrate convergence

toward a common steady-state equilibrium across all tested initial conditions, provided that growth parameters remain balanced. Specifically, after prolonged co-culture periods, initial differences diminish, resulting in nearly identical stable populations for all three strategies (Figure S1c). These findings suggest that, under balanced growth conditions, initial population distributions exert limited influence on long-term outcomes when growth parameters are optimally balanced, highlighting the inherent robustness of the tripartite coexistence to initial perturbations.

Notably, our simulations also reveal a transient ecological advantage for private metabolizers under conditions of elevated initial cheater densities. In scenarios where cheaters initially predominate, their rapid overexploitation of shared resources suppresses public metabolizers, thereby reducing competitive pressure and allowing private populations to expand more rapidly (Figure S1b). While this shift gives rise to short-term deviations from typical community dynamics, long-term equilibria consistently re-establish tripartite coexistence once growth parameters stabilize, highlighting the system's resilience and adaptive capacity.

Collectively, these results highlight that while short-term community dynamics are sensitive to initial conditions and population imbalances, the long-term stability and coexistence of these microbial strategies are primarily determined by balanced growth conditions rather than initial community composition. Such insights are crucial for designing and managing stable synthetic microbial ecosystems capable of sustaining robust performance across a range of starting configurations and environmental contexts.

(c) Impact of the interaction coefficients

The ecological interactions within microbial communities are significantly shaped by parameters α and β, which govern the benefit-sharing and competition dynamics in the system. This study explores how changes in α and β independently affect community dynamics, holding other parameters constant. The current analysis specifically considers a previously balanced community, characterized by a higher intrinsic growth rate of public metabolizers.

A higher α value indicates greater resource exploitation by cheaters from public metabolizers. When α exceeds β, cheater populations grow substantially, often at the expense of public metabolizers due to intense resource diversion. This imbalance can destabilize the community, significantly reducing the population of public metabolizers crucial for ecosystem stability. However, excessively high values of α can ultimately also reduce cheater populations, as their survival directly depends on the availability of public metabolizers. Parameter β quantifies competition between public and private metabolizers for shared nutrients. Higher β values indicate intense competition, severely constraining the growth of both populations. When β significantly exceeds α, the private metabolizer population cannot compete effectively and is driven towards extinction. Conversely, lower β values permit independent growth, but risk unchecked proliferation of private metabolizers, potentially destabilizing the community.

The interactions between α and β critically define coexistence conditions among public, private, and cheater populations. From the analysis and visual inspection of the provided figures, the optimal scenario for stable coexistence is observed when α is moderately higher than β. Under these conditions, cheaters thrive without causing substantial harm to public metabolizers, and public and private populations maintain competitive yet balanced growth. Deviations from this optimal balance, particularly excessively high β or disproportionately high α, disrupt community stability, resulting in dominance or extinction events. Figure 3 illustrates these dynamics, emphasizing parameter ranges that best support stable coexistence among all three microbial populations.

(d) Stability of a Two-Strategy Coexistence

In populations with two co-existing strategies, the third strategy can often invade due to the interplay of exploitation and competition dynamics. For example,when public metabolizer and

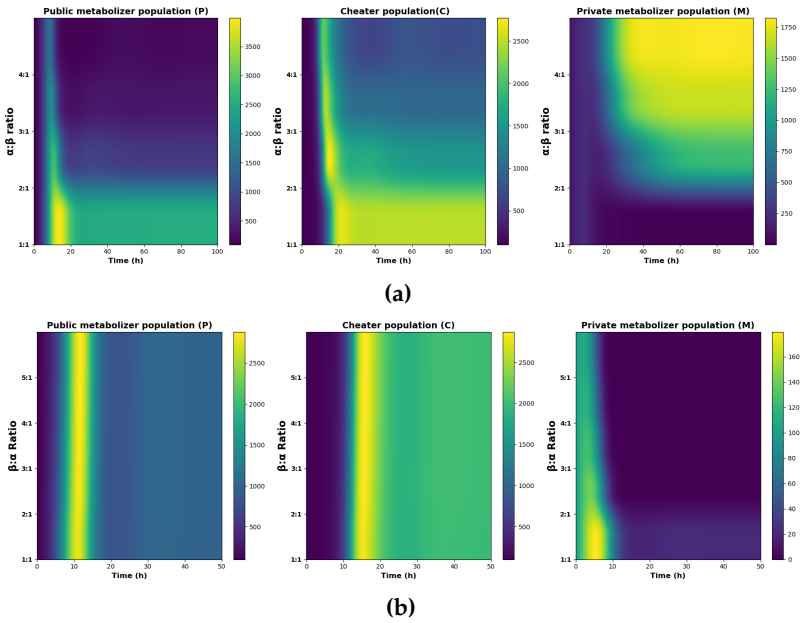

Figure 3. Impact of the interaction coefficients on a stable Population. We consider a high-growth condition for the public metabolizer as indicated in our previous analysis. (a) Higher benefit sharing α enhances the Public metabolizer and cheater abundance, resulting in a higher proportion of private metabolizer. (b) Higher competition loss β causes public metabolizer and cheater populations to survive at minimal density, while the private metabolizer goes extinct.

cheater coexist $(P, C > 0, M = 0)$, the fitness of public is $F_p = r_p - \alpha C - \eta P$, and the fitness of cheater is $F_c = \alpha P - \nu C$. Private metabolizer can invade this system if their fitness, $F_m = r_m - \beta P - \gamma M$, exceeds both F_p and F_c, which occurs when $r_m > r_p - \alpha C$ and $r_m > \alpha P - \nu C$. In coexistence between public metabolizer and private metabolizer $(P, M > 0, C = 0)$, the fitness of P is $F_p = r_p - \eta P - \beta M$, and the fitness of M is $F_m = r_m - \gamma M - \beta P$. Cheaters can invade this system by exploiting public metabolizers, achieving fitness $F_c = \alpha P > F_C$, provided $\alpha > \eta$. Lastly, in a system with cheaters and private metabolizers $(C, M > 0, P = 0)$, where $F_c = \alpha P - \nu C$ and $F_m = r_m - \gamma M - \beta P$, public metabolizers can invade if $r_p > r_m - \beta M + \gamma M$. These mathematical relationships reveal that the coexistence of two strategies is typically unstable, as the third strategy can always exploit competitive or cooperative dynamics to establish itself, necessitating a mixed ESS for population stability.

(e) Stability of a mixed strategy: Coexistence of all three strategies

The dynamics work by increasing the proportion of strategies that yield higher payoffs. For instance, if the Public strategy offers greater fitness compared to private, the population will evolve toward a public-cheater equilibrium. Similarly, if the private strategy offers a higher payoff than the public, the system will shift toward a private equilibrium. However, only cheating is generally discouraged, as mutual cheating leads to negative payoffs. However, in the case of public, cheater, and private mixed equilibrium, the populations have a lower fitness because of a relative loss of cooperation and competition, which reduces the effectiveness of this strategy. Therefore, the system moves toward mixed stable equilibria. The overall dynamics push the population toward a stable state where no strategy dominates, and players have little incentive to switch strategies once equilibrium is reached. We are considering the scenario where all three strategies coexist $(P, C, M > 0)$, and the system reaches a state called a mixed ESS, where no single strategy can invade and dominate. All the payoffs are equal at this ESS.

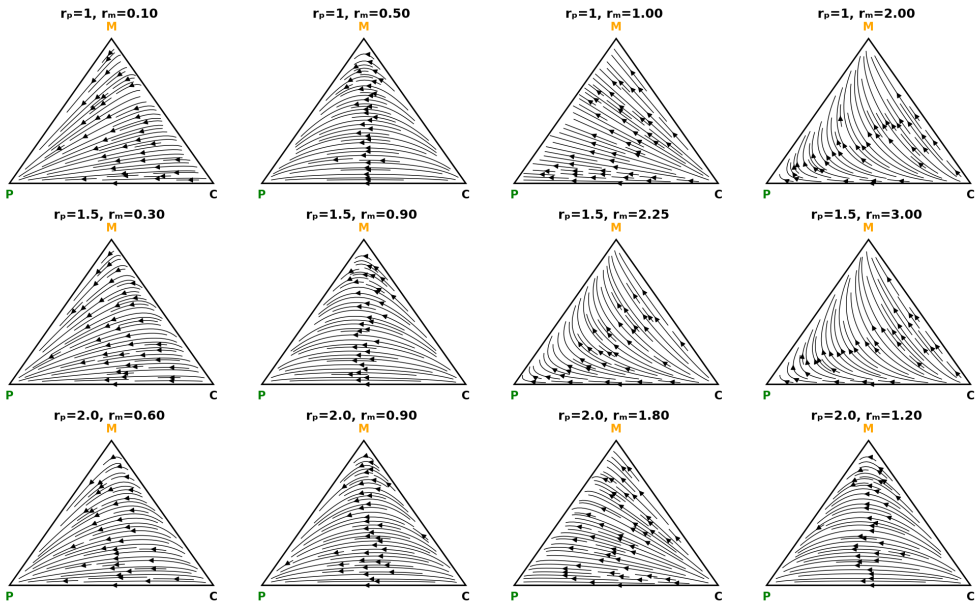

Figure 4. Phase-flow diagrams showing payoff-driven shifts in total population dynamics. Each triangle represents all possible relative abundances of three strategies. Streamlines trace evolutionary trajectories based on our generalized Lotka–Volterra model. As intrinsic growth rates (r_p, r_m) vary, the system shifts from mixed populations to dominance by a single strategy. The flow line converging to a single vertex indicates evolutionary takeover by that strategy. In contrast, if the flow converges to an interior point or along an edge, it indicates a stable mixed population maintained by balanced payoffs.

$$\frac{dP}{dt} = 0, \quad \frac{dC}{dt} = 0, \quad \frac{dM}{dt} = 0.$$

Figure 4 illustrates how total population dynamics evolve under different payoff configurations across the $P-C-M$ simplex. Each triangle shows how the community composition changes over time, with arrows indicating the direction of population flow based on our generalized Lotka–Volterra model. When one strategy has a strong growth advantage, trajectories flow toward the corresponding vertex, leading to complete dominance. In contrast, when payoffs are more balanced, the flow converges toward an interior point or along an edge, representing a stable coexistence of multiple strategies. These patterns reflect how differences in public and private payoffs shape not just relative proportions but the entire structure and size of the microbial community.

The coexistence of all three strategies in the mixed ESS represents a delicate balance governed by intrinsic growth rates, competition, and exploitation dynamics. This mixed strategy is critical for understanding and predicting the long-term dynamics of microbial communities. The equilibrium proportions are:

- proportion of public metabolizer :

$$P^* = \frac{\nu\left(r_p\,\gamma \;-\; \beta\, r_m\right)}{\alpha^2\,\gamma \;+\; \nu\,\eta\,\gamma \;-\; \nu\,\beta^2}$$

- proportion of cheater :

$$C^* = \frac{\alpha\left(r_p\,\gamma \;-\; \beta\, r_m\right)}{\alpha^2\,\gamma \;+\; \nu\,\eta\,\gamma \;-\; \nu\,\beta^2}$$

- proportion of private metabolizer :

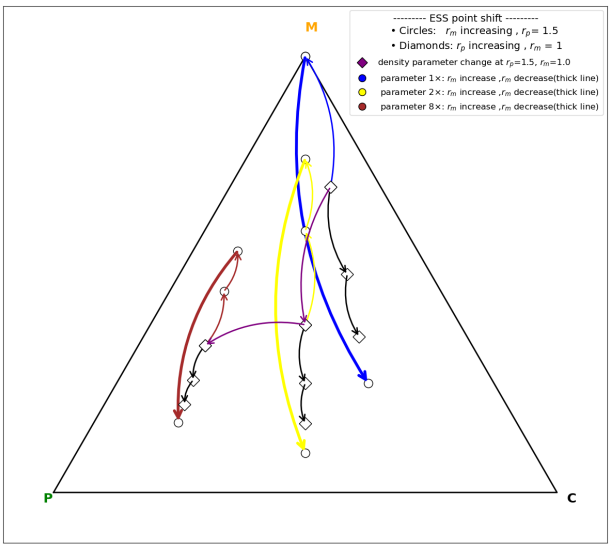

Figure 5. Movement of the Evolutionary Stable Strategy (ESS) across the $P-C-M$ simplex as public (r_p) and private (r_m) metabolizer payoffs are varied. Each point represents a stable population composition ESS, where $P+C+M=1$. Circles indicate increasing r_m with fixed $r_p=1.5$; diamonds indicate increasing r_p with fixed $r_m=1$. The arrows show the direction of the ESS shift under parameter changes. When the density parameter is increased, the ESS shifts toward the $P-M$ edge of the triangle. From this new starting point, increasing r_m pushes the ESS further toward the M corner, while increasing r_p shifts it toward a $P-C$ edge where we still have a coexistence of mixed population.

$$M^* = \frac{\alpha^2 r_m + \eta \nu r_m - \beta \nu r_p}{\alpha^2 \gamma + \nu \eta \gamma - \nu \beta^2}$$

These expressions describe the equilibrium portion where all three strategies coexist in the population.

Increasing the density parameter causes the ESS trajectory to bend toward the $P-M$ edge (Figure 5), reflecting how ecological crowding influences the balance between cooperative and self-sufficient strategies. From this new baseline, further changes in intrinsic growth rates determine the direction of ESS movement. When the growth rates of public r_p and private r_m metabolizers are comparable, the ESS lies inside the triangle, indicating stable coexistence of all three populations. As r_p increases relative to r_m, public metabolizers gain a growth advantage, and the ESS shifts toward the $P-C$ edge, where the public metabolizer or cheater population has domination. Conversely, increasing r_m the relative fitness of public metabolizers, and the ESS moves toward the M edge, where private metabolizers dominate. Once the critical threshold $r_p = \beta r_m/\gamma$ is crossed, public metabolizers can no longer maintain a positive payoff. Their extinction leads to the collapse of cheaters. As a result, the ESS converges to the M apex, where only private metabolizers remain. In summary, when public metabolizers are highly rewarded, stable coexistence of all three is possible. But if private metabolizers gain a similar advantage, evolutionary dynamics drive the system toward complete domination by private metabolizers. These results highlight how payoff asymmetries and density-dependent factors shape long-term evolutionary outcomes in structured microbial communities.

(f) Numerical lab evolution experiment

Under the discussed appropriate ecological conditions (Figure 5), these interactions give rise to stable coexistence among public, cheater, and private strategies. While such mixed equilibria may be associated with reduced average fitness due to the combined costs of

cooperation and exploitation, they represent a dynamic balance in which no single strategy can dominate persistently. Here, the success of each strategy is determined not only by its intrinsic growth rate but also by its interactions with others. To further capture evolutionary realism, we employed laboratory-inspired evolutionary simulations that allowed individual strategies to change between generations, mimicking realistic microbial adaptation. These transitions represent mutation events or phenotypic switching, mechanisms frequently observed in microbial populations adapting to environmental and selective pressures. First, we allowed 30% of private metabolizers to gradually transition (at a rate of 10% per generation) into cheater (Figure S3 a) or public metabolizer (Figure S3 b) strategies. Still, all three strategies are present after seven generations. This mixed population structure remains stable even as the transition rate increases to 60%, demonstrating the system's robustness to evolutionary perturbations. Including such evolutionary flexibility allows the model to probe how mutational plasticity contributes to system stability under shifting ecological landscapes.

Additional simulations (Figure 6)further demonstrate that when public metabolizers gain a substantial growth advantage, doubling the rate of private metabolizers, the system still supports coexistence, though at reduced equilibrium densities. This underscores the fragile ecological balance required to sustain public contributors. Although public metabolizers and cheaters may persist in small fractions, their decline coincides with an increase in total population size. These outcomes highlight the sensitivity of coexistence to fitness asymmetries and demonstrate how imbalances in growth advantages can lead to the collapse of strategic diversity within microbial communities.

4. Discussion and Conclusion

The combined results of numerical simulations and EGT emphasize the critical conditions necessary for stable coexistence among public metabolizers, cheaters, and private metabolizers within microbial communities. Both modelling frameworks consistently reveal that maintaining a growth advantage for public metabolizers is essential for long-term coexistence. When growth rates are balanced and the cost of competition remains lower than the cost of cooperation, our simulations indicate the system tends to stabilize and steady state around a point that includes all three strategies.

However, imbalances in growth rates significantly alter community dynamics, as highlighted by numerical analyses of various initial population ratios and growth rate parameters (Figure 2,Figure S1). A higher initial ratio of cheaters consistently showed detrimental impacts on public metabolizers, suppressing their maximum population densities and long-term survival. In contrast, private metabolizers benefited indirectly under these conditions, achieving higher equilibrium densities because of decreased competition from the declining public population. These findings align with evolutionary game-theoretic predictions, supporting the conclusion that cooperative strategies are especially vulnerable to exploitation unless protected by stabilizing conditions.

Further insights emerged from evolutionary game simulations that examined population shift strategies and provided valuable insights into resilience and adaptability within microbial communities. Even when a large fraction of private metabolizers transitioned to cheating or public strategies, the system often retained all three populations, indicating a robust tolerance to evolutionary changes. Notably, even with a large fraction of private metabolizers mutating toward other strategies, it remains possible for all three strategies to persist within the community during the evolutionary phases. However, substantial shifts toward cheaters imposed significant pressure on public metabolizers, often leading to their elimination unless cooperation costs remained sufficiently low and ecological competition was constrained.

An extensive exploration of parameter spaces, including the effects of varying $\alpha : \beta$ and $\beta : \alpha$ ratios, reinforced these findings. Specifically, higher $\alpha : \beta$ ratios, representing elevated exploitation by cheaters relative to competition costs, consistently reduced the stability of cooperative interactions. Conversely, optimizing the balance between these parameters promoted conditions

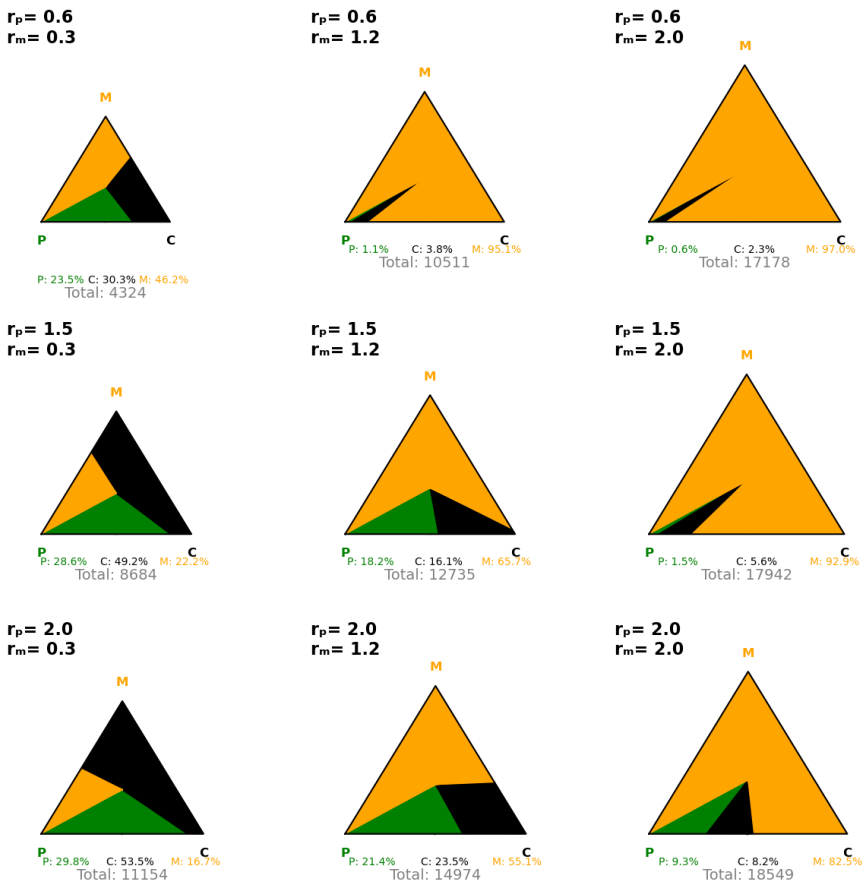

Figure 6. Evolutionary outcomes of a three-member three-strategy microbial community under varying payoff asymmetries in heterotrophic microbial consortia. Each triangle depicts the final population distribution of public, cheater, and private metabolizers of each generation. Colour regions indicate their relative proportions: green for P, black for C, and orange for M; numeric labels show final percentages and total population size. When $r_p > r_m$, public metabolizers hold a growth advantage, cheaters exploit this benefit, and all three strategies coexist with a low total population. As r_m increases and surpasses r_p, M dominates and total population increases, while P and C persist only in small fractions.

favourable for sustained coexistence. These findings highlight how the intensity and direction of competitive and exploitative pressures shape the long-term evolutionary landscape of microbial communities. Moreover, results from both numerical simulations and evolutionary game analysis revealed that the public growth rate should be higher than the private growth rate but should not exceed it by more than a factor of 1.5 to 2 (Figure 6, Figure 2). This narrow window defines the optimal condition for maximizing overall community density while preserving diversity. If the public growth rate is double that of the private rate, the mixed community achieves maximum density. However, extreme imbalances—where one strategy significantly outpaces the others—lead to competitive exclusion. Elevated public growth results in the dominance of public metabolizers and cheaters, while significantly higher private growth leads to the extinction of the other two populations (Figure 6, Figure 2).

Whilst most studies on cooperation and defection dynamics primarily focus on two-player, two-strategy games, sometimes extend to a three-strategy rock-paper-scissors framework [13,32, 39], in this work we systematically investigated a three-player three-strategy game.

Using EGT, we confirmed that neither pure nor mixed strategies involving only two community members were evolutionarily stable (Figure 4, Figure 5). Only the three-strategy configuration met the criteria for an ESS, emphasizing that long-term microbial stability depends on the coexistence of cooperative, exploitative, and self-sufficient strategies. This outcome reveals the structural interdependence of functional roles in microbial ecosystems and demonstrates why diversity is not merely beneficial but evolutionarily necessary.

Finally, these results provide both mechanistic insight and practical guidance for managing microbial communities. Stable coexistence is contingent upon a precise interplay between growth advantages, cooperation costs, and population structure. Our findings suggest that targeted interventions, such as modulating nutrient levels, introducing feedback-limited gene circuits, or engineering metabolic constraints, can reinforce this balance and sustain desired community states. By identifying parameter regimes that support robustness and diversity, this work contributes to the rational design of synthetic consortia and the ecological stabilization of natural microbial systems. These insights extend beyond theoretical ecology. They inform real-world applications in biotechnology, environmental management, and microbiome therapeutics, where maintaining functional diversity and preventing collapse are essential to community performance. Understanding the ecological logic behind coexistence enables predictive control over microbial behaviour, advancing both the engineering and conservation of complex biological systems.

Code availability

Model development and numerical analysis have been performed in Python, and all code is available under: `https://github.com/Computational-Biology-Aachen/FightClub/`. In particular, with JupyterNotebook main.ipynb and simplex-code.ipynb, all figures from this manuscript can be reproduced.

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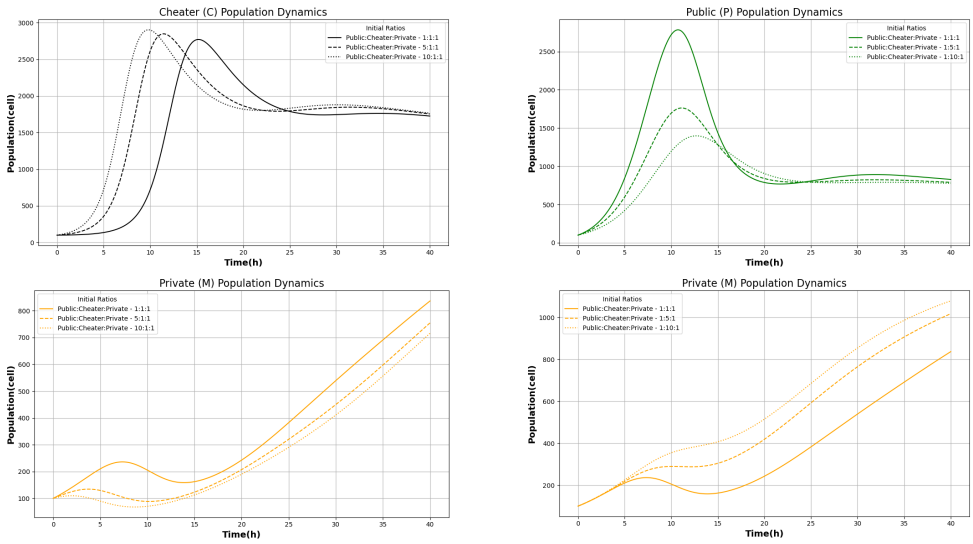
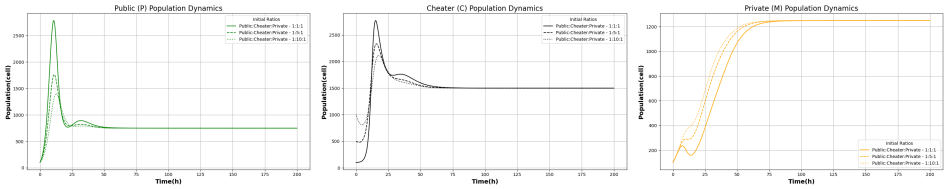
Supplementary Material



(a) Different ratios of the public metabolizer in the initial population density

(b) Different ratio of the cheater in the initial population density



(c) Long-term co-culture with different initial ratios of all three populations

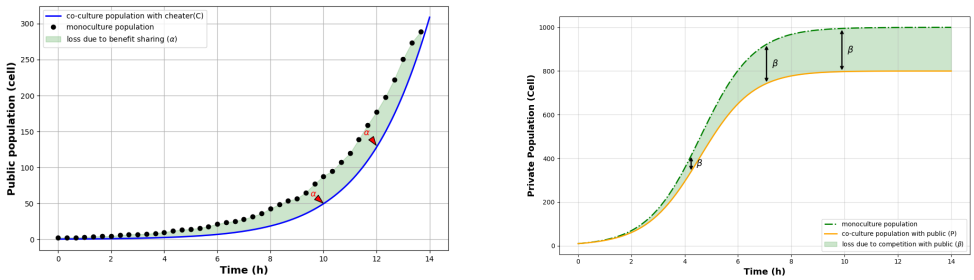
Figure S1. The population dynamics of all three community members depend on the initial ratios. (a) and (b) illustrate how early dynamics differ with changing initial abundances of the public metabolizer and the cheater, respectively. (c) shows that despite these initial differences, all communities converge to a similar steady state in long-term coculture. This indicates that long-term coexistence is largely independent of the initial population composition.

A. Parameters sensitivity analysis

In microbial ecosystems, such sensitivity analyses are essential to identify which biological interactions, such as cooperation or competition, most critically influence population stability and composition. The relative influence of each parameter on system behaviour can be quantified by introducing perturbations to key parameters, such as α, and examining the resulting changes in steady-state population densities. To assess the relative impact of a given parameter on a population ($X \in \{P, C, M\}$), the deviation between the perturbed and baseline population trajectories is integrated over time, as expressed by the following relation:

$$\text{Relative Impact} = \frac{\sum_t (X_{\text{perturbed}}(t) - X(t))}{\sum_t X(t)}$$

where $X_{\text{perturbed}}(t)$ denotes the population trajectory under parameter perturbation, and $X(t)$ represents the steadystate baseline trajectory. This formulation captures the cumulative deviation caused by the perturbation, offering a time-integrated measure of sensitivity over the simulation



(a) Estimating the cooperation parameter (α) **(b)** Estimating the competition parameter (β)

Figure S2. Estimating interaction parameters from mono- and co-cultures. (a) To estimate the cooperation parameter (α), we compare the growth trajectory of the public metabolizer (P) in monoculture (black points) with its growth in co-culture with the cheater (C) (blue line). The area between the curves (shaded green) quantifies the reduction in population size due to shared public goods, highlighting the fitness cost of exploitation. (b) To estimate the competition parameter(β), we compare the growth of the private metabolizer (M) in monoculture (green dashed line) and co-culture with the public metabolizer (P) (orange line). The shaded area illustrates the population loss attributable to indirect competition for sucrose resources. These pairwise measurements enable data-informed parametrization of our mathematical model.

(a) Shift from private to cheater population **(b)** Shift from private to public population

Figure S3. Population dynamics during evolutionary shifts from private to cheater and public strategies. (a, b) Population dynamics illustrating an initial shift of 30% of private metabolizers transitioning gradually into either cheater (a) or public metabolizer (b) strategies, with subsequent increases of 10% each generation. All three strategies remain coexistent and stable, even when up to 60% of the private metabolizer population has transitioned, demonstrating resilience and ecological stability under evolutionary changes.

period. As parameter sensitivity may vary depending on the magnitude of perturbation, the average impact is computed across multiple perturbation steps:

$$\text{Average Impact} = \frac{1}{N}\sum_{i=1}^{N}\text{Relative Impact}_i$$

Here, N is the total number of perturbation steps. Averaging across these steps helps reduce the influence of noise or isolated fluctuations and provides a more robust estimate of each parameter's influence. This approach provides a robust measure of a parameter's influence, minimising the effects of noise or extreme variations. To ensure consistency and comparability across different parameters and populations, the computed impacts are normalised to a standard scale within the range $[-1, 1]$.

$$\text{Normalized Impact} = \frac{\text{Impact} - \min(\text{Impact})}{\max(\text{Impact}) - \min(\text{Impact})}$$

Normalisation allows for direct comparison between parameters, ensuring that those with stronger relative effects are identified, even when absolute magnitudes differ due to different units or scales.

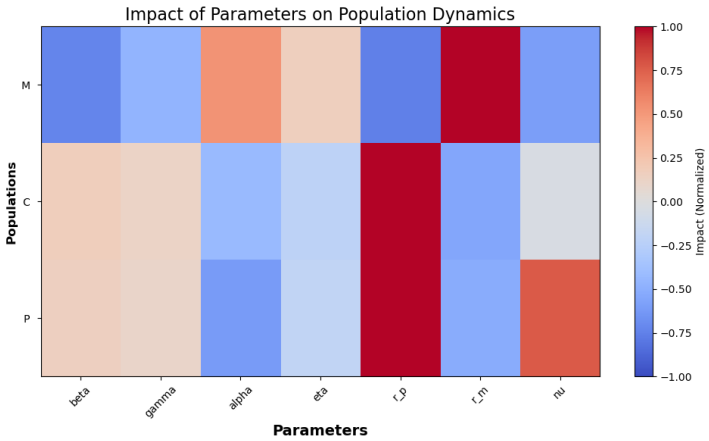


Figure S4. Impact of competition (β), cooperative (α), growth rate (r_p, r_m), and density-dependent constraints (ν, η, γ) parameters in P,$C$and M population dynamics. This heat map shows which parameters have a positive or negative influence on population dynamics of the tripartite co-culture system.